

ANISH DHAR

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PROFESSIONAL SUMMARY

Mechanical engineer with a strong background in structural and thermal simulation, precision manufacturing, and end to end product development. I have a proven track record in fluid and mechanical systems, notably taking full ownership of complex assemblies and novel ducting and joining systems from the initial concept all the way to manufacturing release. I am proficient in 3D modeling tools like Autodesk Inventor and SolidWorks, as well as GD&T using ASME Y14 and ANSI Y14 standards, and design validation techniques like DVPR. I possess a strong digital mindset with hands on experience in AI tools such as Physics Informed Neural Networks and CFD/FEA analysis. I consistently maintain clear communication with internal stakeholders and ensure engineering projects are executed under strict requirements.

EXPERIENCE

Engineering Trainee, New Product Development | Glatt Systems Pvt. Ltd., Pune, IN Jun 2025 to Present

- Designed a novel SS316 ducting joining system to eliminate legacy flanged connections. I created 3D models and drawings, and successfully qualified the prototype at 2 bar pressure in a single connection acceptance test as part of the DVPR process.
- Took complete design ownership of a first of its kind containment bin washing solution for highly potent APIs, valued at INR 3.5 Cr. The structure stood over 9 meters tall, was partially embedded in the ground with specialized containment valves, and was designed to wash bins ranging from 600 to 2500 liters in size. I delivered the full design package and BOMs from scratch while coordinating both procurement and manufacturing.
- Took design ownership of a new lab scale extruder spheronizer. I guided the product through its complete development cycle, covering everything from mechanical CAD and manufacturing to functional testing and presenting the machine at industry exhibitions.
- Managed Change Requests and Change Orders across active design packages. I tracked revision controlled drawing releases through Vault PDM, maintaining clear and credible communication with vendors and the shop floor team.
- Incorporated shop floor feedback to improve our manufacturing processes, including machining, sheet metal, and joining. I maintained strict GD&T drawing standards based on ASME Y14, along with clear engineering calculations and tolerance stack up documentation throughout the fabrication stage.
- Contributed structural design expertise to a 3 story all steel vertical granulation line. I collaborated closely across mechanical and process sub teams to ensure internal customer satisfaction through thorough pre production validation.

Engineering Intern, Thermal Systems | Saarloha Advanced Materials Pvt. Ltd., Pune, IN Jun 2024 to Aug 2024

- Surveyed the water cooling circuit of a 35T Electric Arc Furnace and continuous casting line. I used analytical thinking and plant instrumentation data to identify thermal loss pathways caused by slag deposition on the cooling panels.
- Delivered a structured improvement report proposing design modifications to reduce parasitic heat losses in the EAF water cooling loop, aligning directly with the site's green manufacturing targets.

ENGINEERING PROJECTS & COMPETITIONS

Thermal Systems, Electronics, SAE e-BAJA & ATVC | VJTI Racing Jan 2022 to Mar 2024

- Developed low loss manifold geometry to reduce static pressure drop across the forced air distribution network, which decreased the auxiliary battery load. Our team achieved AIR 1 in the Sales Presentation and AIR 5 Overall at SAE e-BAJA 2024.

- Designed a jet impingement cooling enclosure for the motor controller using SLA and FDM additive manufacturing. This design targeted a 25°C reduction in peak operating temperature compared to the baseline air cooled configuration.
- Led the low and high voltage system design for our ISIE IKR 2023 EV entry. I implemented a multi layer startup interlock sequence that helped us win the Best Accelerating Vehicle award and place AIR 5 in the EV category.

Converging Diverging Nozzle, IPA and Air Rocket Engine | *Personal Project*

- Designed a C-D nozzle for 300 N sea level thrust. I handled the CNC machining of the mild steel nozzle on a Jyoti VMC850, taking care of the fixturing design, toolpath programming, and post test failure analysis using DFMEA concepts.

Numerical Investigation of Lattice Based Cooling Structures | *Capstone, VJTI*

Apr 2025

- Developed and meshed lattice unit cell geometries in ANSYS Fluent. I conducted mesh convergence studies to better understand convective heat transfer enhancements and pressure drop trade offs across various candidate topologies.

Compressive Behavior of Vertically Graded Lattice Structures | *Honours, VJTI*

May 2025

- Modeled graded density lattice structures under uniaxial compression using ANSYS Mechanical. I extracted load displacement curves via structural FEA to accurately quantify their energy absorption.

EDUCATION

B.Tech in Mechanical Engineering | *VJTI Mumbai* | CGPA: 7.67

Dec 2021 to May 2025

Relevant coursework: Thermodynamics, Fluid Mechanics and CFD, Material Sciences, Manufacturing, Solid Mechanics, Additive Manufacturing.

LICENSES & CERTIFICATIONS

Applied Computational Fluid Dynamics, Siemens

Apr 2026

Cleanroom Fundamentals & Semiconductor Technologies, University at Buffalo

Apr 2026

SOLIDWORKS: Design Study and Optimization, LinkedIn Learning

Jun 2023

TECHNICAL SKILLS

CAD and PDM: Autodesk Inventor for 3D modelling, large assemblies, and sheet metal, SolidWorks, Vault PDM, Fusion 360

Simulation and AI Tools: ANSYS Mechanical for structural FEA, ANSYS Fluent for CFD, MATLAB and Simulink, Python for AI tools, PINN, and SPH simulation

Engineering Standards and Methods: GD&T using ASME Y14 and ANSI Y14, Tolerance Stack up, DFMEA, DVPR, Change Requests and Orders, BOM Management, Requirements Review

Manufacturing Processes: Machining using CNC Jyoti VMC850, Sheet Metal, Joining, FDM and SLA Additive Manufacturing, Fixturing